

MERCHANTLENS

AI Representation Optimizer for E-Commerce

PRODUCT DOCUMENT

Kasparro Lab Hackathon — Track 5 (Advanced)

Stack

Next.js 16 · TypeScript · Tailwind CSS
Groq API · LLaMA 3.3 70B

01 THE PROBLEM

AI shopping agents — ChatGPT Shopping, Perplexity, Google SGE — are reshaping how users discover products. Instead of browsing, they ask questions like "What are the best sustainable shoes under \$100?" and receive AI-curated answers.

These systems depend entirely on merchant-provided data: product descriptions, FAQs, policies, structured markup, and trust signals. When that data is incomplete or ambiguous, the AI either skips the merchant or misrepresents them — costing sales.

Many SMB merchants lack structured product data	Weaker AI visibility	Limited AI readiness tooling for AI readiness diagnostics	\$25B AI commerce market projected by 2028
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Most merchants optimise for SEO and human shoppers. There are only a few tools for AI representation optimization.

02 OUR SOLUTION

MerchantLens is a diagnostic platform that shows Shopify merchants exactly how AI agents perceive their stores — and provides a ranked action plan to close the gap between AI perception and merchant intent.

Capability	Description
AI Readiness Score	Five-dimension scoring: product clarity, FAQ coverage, trust signals, policy completeness, structured data.
Top Issues Dashboard	Severity-ranked issues (High / Med / Low) with plain-language descriptions and root-cause context.
Ranked Action Plan	Prioritised recommendations ordered by impact-to-effort ratio — not just a list, a playbook.
AI Perception View	Full narrative of how AI agents currently read the store, plus perceived

	strengths and weaknesses.
Perception vs. Intent	Side-by-side gap analysis showing where AI understanding diverges from merchant positioning.
Multi-Format Input	Accepts store URLs, pasted product descriptions, and uploaded policy documents.

03 CORE USER JOURNEY

Capability	Description
Step 1 — Input	Merchant pastes a product description, submits a store URL, or uploads a policy document.
Step 2 — Validation	System validates that all inputs belong to the same merchant context before unified analysis begins.
Step 3 — Analysis	LLM evaluates trust signals, FAQ quality, product clarity, structured data presence, and policy completeness.
Step 4 — Dashboard	Merchant receives a scored, tabbed diagnostic report with issues, recommendations, AI view, and comparison.
Step 5 — Optimization	Built-in toolkit: Schema.org validator, meta tag analyzer, AI readiness checklist, and export to report.

04 KEY PRODUCT DECISIONS

Diagnostic layer, not a full agent

We scoped down from a full AI agent simulation. The core merchant problem was visibility — diagnostics are more actionable, demoable, and directly match the challenge.

Product clarity over AI complexity

We avoided vector DBs, multi-agent pipelines, and fine-tuning. Structured prompts with strict JSON schemas produced more reliable, explainable outputs that merchants can act on.

Consistency validation before analysis

Input validation prevents conflicting or unrelated content from generating misleading unified reports — improving trust in the output.

Multi-format input

Merchant data lives in URLs, descriptions, and documents. Supporting all three made the tool feel closer to real merchant workflows.

No Shopify API in MVP

Deep Shopify integration would shift focus from diagnostics to auth and sync flows. URL-based input gives 80% of the value with 20% of the complexity.

No authentication in MVP

Authentication did not improve core user value. We invested the saved time in analysis quality and dashboard polish instead.

05 TRADEOFFS

Tradeoff	What We Chose	Why
Breadth vs. Depth	Fewer, polished workflows	Coherent merchant experience over scattered AI utilities
Real-time vs. Speed	Faster iteration, URL input	Shopify API integration would slow development without improving core value
Technical complexity vs. UX	Simple AI architecture	Explainable outputs, faster prompt iteration, stronger merchant usability

06 TARGET USERS

Primary User

Small-to-mid-sized Shopify merchants, particularly D2C brands with content-heavy product pages who rely on product discovery and care about conversion trust.

Merchants who:

- Optimise for SEO but have zero visibility into AI perception
- Write product descriptions manually without AI-agent context
- Lack a feedback loop on what AI systems understand vs. miss
- Want better placement in AI-assisted shopping recommendations

Current Gap

Exists Today	Missing
SEO tools	AI readiness audit
Ad analytics	AI perception score
CRO platforms	Agent gap analysis
Social tools	Structured data check

07 TECHNICAL ARCHITECTURE

Capability	Description
Frontend	Next.js 16 (App Router), TypeScript, Tailwind CSS — server components with edge-compatible API routes.
AI Engine	Groq API with LLaMA 3.3 70B — strict JSON schema prompting ensures structured, parseable diagnostic outputs.
Analysis Pipeline	System prompt defines a five-dimension scoring rubric. Output is validated, parsed, and mapped to dashboard components.
Input Handling	Store URLs, product descriptions (up to 5,000 chars), and uploaded policy documents (.docx, .pdf, .txt).
State & History	Client-side session state with auto-save to analysis history. One-click export to structured .txt report.

08 FINAL OUTCOME

MerchantLens demonstrates that AI representation optimization is a real and underserved merchant problem. The platform delivers:

- Visibility into AI perception — not a guess, a scored diagnostic
- Structured readiness report across 5 critical dimensions
- Trust signal analysis with severity-ranked issue cards
- Ranked action plan ordered by impact-to-effort ratio
- Perception vs. intent gap analysis for brand positioning
- Merchant-friendly UX with zero infrastructure overhead

Our goal was not to build the most technically complex AI system. Our goal was to build a thoughtful, usable, and practical AI commerce optimization product that creates real value for real merchants.